

FLIN FLON, MB, Canada

WMO: 719540

Lat: 54.680N

Lon: 101.680W

Elev: 997

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-27.5	-39.6	0.6	-31.7	-34.9	0.8	-27.1	19.1	2.3	17.3	1.0	2.9	50	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	82.3	64.7	79.1	63.3	76.3	61.9	67.8	77.3	65.9	75.1	64.2	72.7	8.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	94.1	72.1	62.5	87.9	70.0	60.8	82.6	68.4	32.8	77.3	31.3	75.2	29.9	72.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.3	14.2	DB	-38.3	88.8	5.3	2.4	-42.1	90.6	-45.2	92.0	-48.2	93.4	-52.0	95.1
			WB	-38.7	72.1	5.2	3.1	-42.5	74.3	-45.6	76.1	-48.5	77.8	-52.3	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	-3.4	1.2	14.5	32.5	46.8	59.3	65.3	62.8	52.2	37.5	18.2	2.2
	DBStd	26.52	14.49	13.43	14.45	11.54	8.78	6.79	5.51	6.04	8.02	8.07	12.09	14.36
	HDD50	7727	1655	1368	1100	527	167	9	0	1	68	396	955	1481
	HDD65	11978	2120	1788	1565	974	566	191	65	114	390	854	1405	1946
	CDD50	1370	0	0	0	3	67	289	473	398	133	7	0	0
	CDD65	145	0	0	0	0	1	21	73	45	5	0	0	0
	CDH74	1137	0	0	0	0	38	198	540	319	42	0	0	0
	CDH80	197	0	0	0	0	6	35	93	58	5	0	0	0
(14) Wind	WSAvg	6.3	5.5	5.7	6.0	6.2	6.6	6.5	6.5	6.8	7.0	7.1	6.7	5.5
(15) Precipitation	PrecAvg	18.5	0.7	0.6	0.7	1.0	1.5	2.6	3.3	2.4	2.4	1.4	1.0	0.9
	PrecMax	26.1	1.4	1.3	1.3	1.9	3.2	4.9	5.1	5.5	4.1	3.0	3.0	1.9
	PrecMin	15.1	0.2	0.0	0.1	0.1	0.2	0.9	1.5	1.0	0.5	0.2	0.4	0.1
	PrecStd	3.0	0.4	0.3	0.4	0.5	0.8	1.1	1.0	1.4	1.1	0.6	0.7	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	38.6	50.1	67.0	78.8	84.2	86.1	85.6	78.5	63.6	46.0	34.2
		MCWB	33.5	33.9	40.4	50.1	57.3	63.9	67.1	67.2	62.6	52.2	39.3	31.8
	2%	DB	28.4	32.6	44.7	59.9	72.6	79.0	82.6	81.3	73.6	57.1	40.6	28.8
		MCWB	26.7	30.3	37.5	46.9	54.6	61.4	65.2	65.2	60.7	49.6	36.6	27.3
	5%	DB	23.2	27.6	39.3	54.3	68.0	75.5	79.6	77.6	68.8	52.9	36.8	25.4
		MCWB	22.3	25.8	34.1	43.5	52.6	59.8	64.0	63.6	58.2	46.4	33.9	24.3
	10%	DB	18.7	21.9	35.1	49.5	63.2	72.3	76.9	74.0	64.8	49.0	33.7	22.0
		MCWB	17.9	20.5	31.8	40.2	49.9	58.3	62.7	62.0	55.8	44.0	31.7	21.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	34.5	40.9	51.4	60.2	68.4	70.9	69.9	65.4	54.5	40.6	31.8
		MCDB	35.4	37.6	49.4	64.5	74.5	77.7	80.6	80.4	74.1	60.8	44.0	34.1
	2%	WB	26.8	30.3	37.5	47.4	56.4	65.2	68.2	67.4	62.0	49.7	36.8	27.5
		MCDB	28.2	32.4	43.9	58.6	68.6	73.7	77.5	77.0	71.2	55.5	39.9	28.7
	5%	WB	22.4	25.7	34.8	44.0	53.8	62.6	66.4	65.5	59.2	47.1	34.1	24.4
		MCDB	23.1	27.6	38.9	53.3	65.2	71.2	76.1	74.1	67.7	52.3	36.7	25.5
	10%	WB	17.9	20.6	31.9	40.7	51.1	60.3	64.6	63.7	56.6	44.4	31.8	21.3
		MCDB	18.8	22.0	34.8	48.7	61.5	68.8	73.6	71.6	63.6	48.4	33.6	22.0
(35) Mean Daily Temperature Range	MDBR		16.1	20.3	23.0	21.8	20.8	18.5	17.9	16.6	15.8	11.8	11.8	14.2
	5% DB	MCDBR	18.4	21.4	24.1	27.4	28.9	24.1	22.1	22.1	22.3	18.7	13.7	16.6
		MCWBR	17.4	18.6	17.5	16.4	14.3	10.4	9.5	10.1	12.2	11.8	10.4	15.7
	5% WB	MCDBR	18.0	20.5	22.4	26.6	25.2	19.8	19.7	18.6	20.9	17.5	12.8	16.4
		MCWBR	17.1	18.0	16.4	16.2	13.5	10.1	10.0	10.3	12.2	11.5	10.2	15.5
(40) Clear-Sky Solar Irradiance	taub		0.237	0.247	0.270	0.329	0.339	0.368	0.390	0.356	0.318	0.289	0.248	0.221
	taud		2.300	2.333	2.356	2.387	2.460	2.401	2.385	2.481	2.577	2.586	2.441	2.306
	Ebn at Noon		235	275	292	285	288	279	270	274	273	259	233	218
	Edh at Noon		17	23	29	33	32	35	35	30	24	19	15	13
(44) All-Sky Solar Radiation	RadAvg		275	626	1110	1586	1751	1766	1769	1448	991	538	252	178
	RadStd		29	45	96	109	181	138	114	111	114	76	27	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+3.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page